TEAM NOISE · HOW DOES THE AUDIO MODEL BEHAVE?

Can an Audio AI Recognize Animals When Sounds Get Noisy or Mixed?

Testing model behavior across clean audio, background noise, mixed sounds, and mimicry

1. HOW WE WORKED WITH THE MODEL

We used a pre-trained sound classifier (AST) to recognize animal sounds. We kept the model closed and stress-tested it by adding static noise, mixing in competing sounds, and testing mimicry.

2. HOW THE MODEL BEHAVED

On clean audio, it's **99% accurate**. But while static noise just makes it unsure, **distinct sounds (sirens, crying babies) completely hijack the model**, stealing its prediction.

3. HOW LLM VISUALIZATIONS HELPED

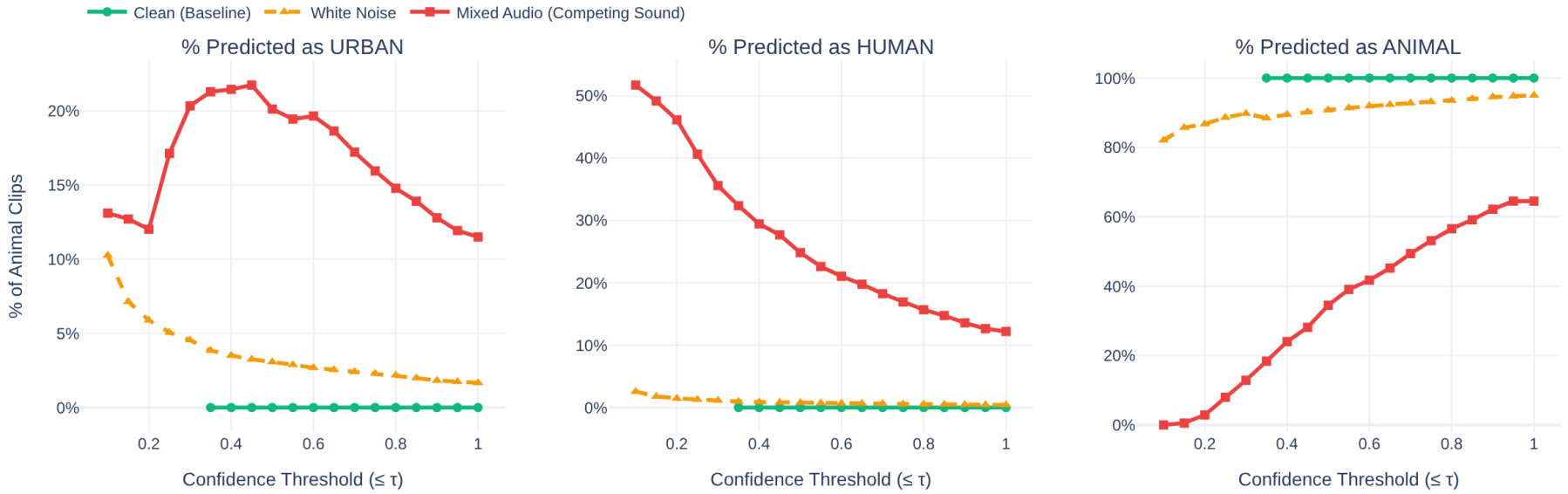
The LLM suggested **3 targeted visualizations** to track confidence loss, compare five sound datasets, and zoom in on mistakes so we could see exactly when and why the model failed.

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VISUALIZATION 1 CONFIDENCE SWEEP

Static Background Noise vs. Competing Mixed Sounds

Tracking what the model predicts as its confidence drops across three sound conditions



- **Clean Audio (Green)**

Flat line at **100% animal predictions** across all thresholds. Pristine animal sounds are never confused with city or human noises.

- **Static Noise (Yellow)**

Static lowers the model's confidence, but **82% to 95% of clips still predict the animal**. Unstructured noise rarely tricks the model into guessing a city sound.

- **Mixed Sounds (Red)**

Distinct background sounds completely overpower the animal: low-confidence clips jump to **>22% City sounds (sirens)** and **~52% Human sounds (crying baby)**.

What This Shows About Model Behavior: Static noise just makes the model less certain, but **competing sounds hijack its prediction** and erase the animal identity.

LLM Role: Suggested plotting predictions across confidence thresholds to expose this takeover.

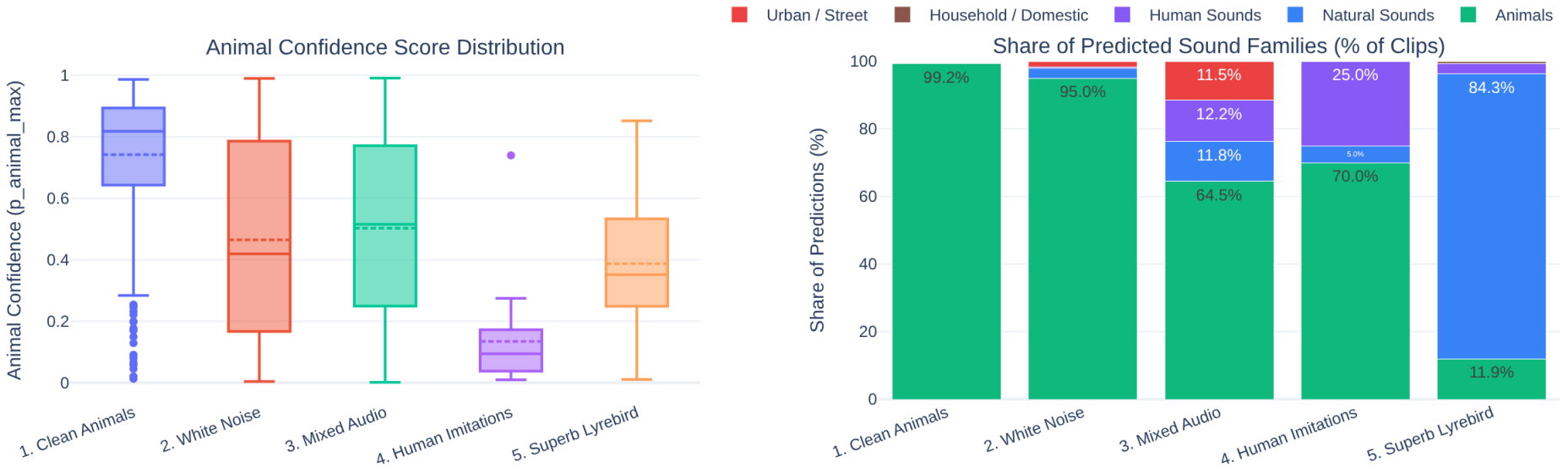
VISUALIZATION 2 CROSS-DATASET VIEW

⏱️ 0:45 – 1:15 (30s) · Slide 3/5

Confidence vs. Actual Guesses Across 5 Sound Datasets

Comparing model confidence (left) with predicted sound categories (right) across 5 scenarios

Cross-Dataset Comparison: Clean Animals vs. Perturbed Audio vs. Imitations & Lyrebird



- **Clean vs. Static vs. Mixed**

Static noise drops confidence but keeps **95% animal predictions**. In mixed audio, animal guesses fall to **64.5%**, splitting into city, human, and nature sounds.

- **Humans Mimicking Animals**

On paper, **70% are labeled as animals**, but the left boxplot reveals that **confidence completely collapsed to near zero (~0.10)**. 25% are detected as human.

- **Superb Lyrebird (Wild Mimic)**

84% predicted as natural bird song. However, 12% predicted as animals because the bird genuinely mimics them (the model accurately detects a frog in clip `XC738172`).

What This Shows About Model Behavior: Looking only at category predictions is misleading—pairing category shares with confidence scores reveals when the model is merely guessing.

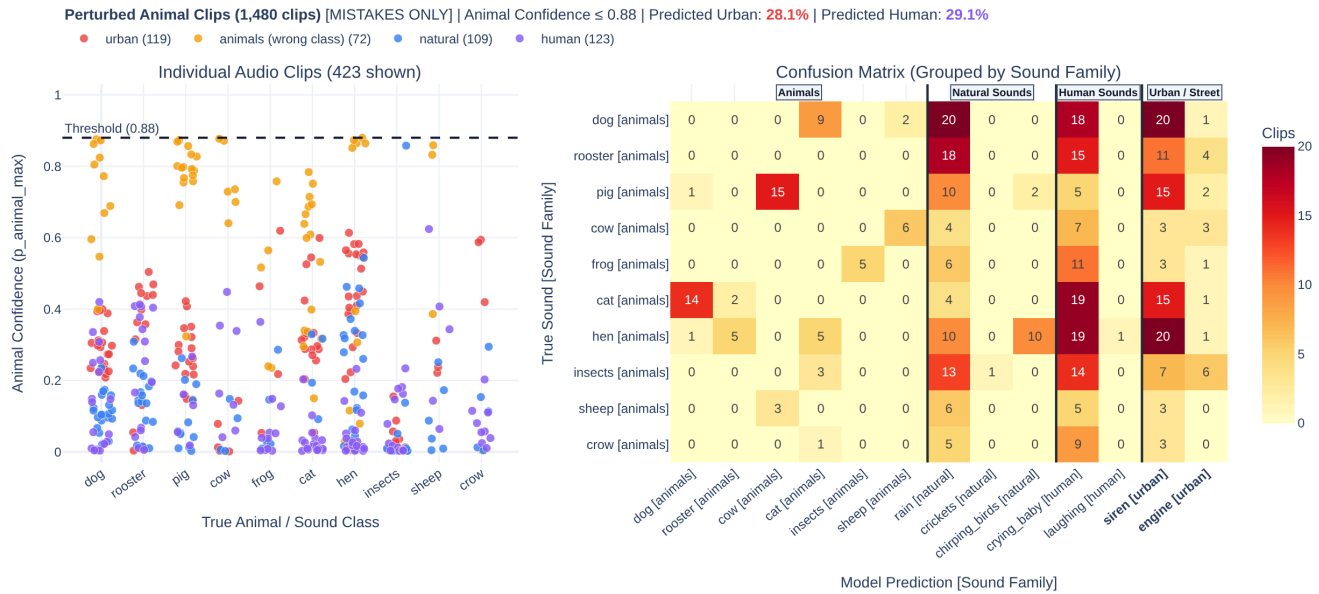
LLM Role: Proposed pairing confidence boxplots side-by-side with stacked category bars.

VISUALIZATION 3 | MISTAKE EXPLORER

⏱️ 1:15 – 1:45 (30s) · Slide 4/5

Mistake Explorer: Why Does the Model Fail?

Zooming in on mistakes only: 443 of 455 mixing errors are the background sound winning



● 97% Are Background Takeovers

Out of 455 mixing mistakes, **443 simply name the competing background sound** (crying baby, siren, rain). The model rarely makes random third-class errors.

● Engine Noise Is Ignored

While sirens and crying babies easily overpower the animal sound, **engine noise is ignored 91% of the time**—the model still successfully recognizes the animal.

● Hen Is Extremely Vulnerable

In animal-on-animal mixes, **hen loses to rooster in all 20 out of 20 tests**—even when the hen is **+10 dB louder**! Rooster survives half of the time.

What This Shows About Model Behavior: Mistakes are not random: loud, distinct sounds (sirens, crying babies, roosters) aggressively drown out softer sounds (hens, barks).

LLM Role: Filtered out correct answers so errors and competing sound takeovers stand out instantly.

AI4VIS REFLECTION PHASE II NEXT STEPS

How the LLM Helped Us Understand the Model

Summary of human–AI collaboration in Phase 1 and our roadmap for Phase 2

WORKING WITH THE LLM (AI4VIS)

LLM as Visualization Partner & Critic

• **Suggested Core Visualizations:** The LLM proposed tracking confidence loss, comparing all five datasets side-by-side, and isolating mistakes.

• **Adversarial Data Checking:** We asked the LLM to audit our summaries against raw predictions; it proved that 97% of mixing errors were simply the distractor winning.

• **Where Human Judgment Was Essential:** The LLM suggested dropping weak data; we refined it into interactive sliders. We also listened to raw audio clips ourselves to confirm mimicry.

BOTH PAIRS CONVERGED & NEXT STEPS

Hen Vulnerability & Opening the Model

• **Both Pairs Independently Agreed:** Both the Human pair and AI pair discovered that **hen** is the model's main weak point (fails on clean data and loses 20/20 mixes to rooster).

• **Phase 2 (Opening the Black Box):** In the next phase, we look inside the model to see *why* loud sirens and roosters steal attention away from animal sounds.

• **Role Swap:** Gurmehak & Elif become the AI-assisted pair; Asiya & Rachid switch to the human-only pipeline.

Key Takeaway: Clean benchmark scores are deceiving. Stress-testing with noise and competing audio—guided by LLM visualizations—uncovered how and why the model fails.

Thank You! · Team Noise